

CHE 396 Quiz 3

Zainab Ahmed

TOTAL POINTS

75 / 100

QUESTION 1

11 40 / 50

- **0 pts** Correct
- **50 pts** wrong
- **20 pts** Partially correct (major mistake)
- ✓ - **10 pts** Partially correct (minor mistake)

QUESTION 2

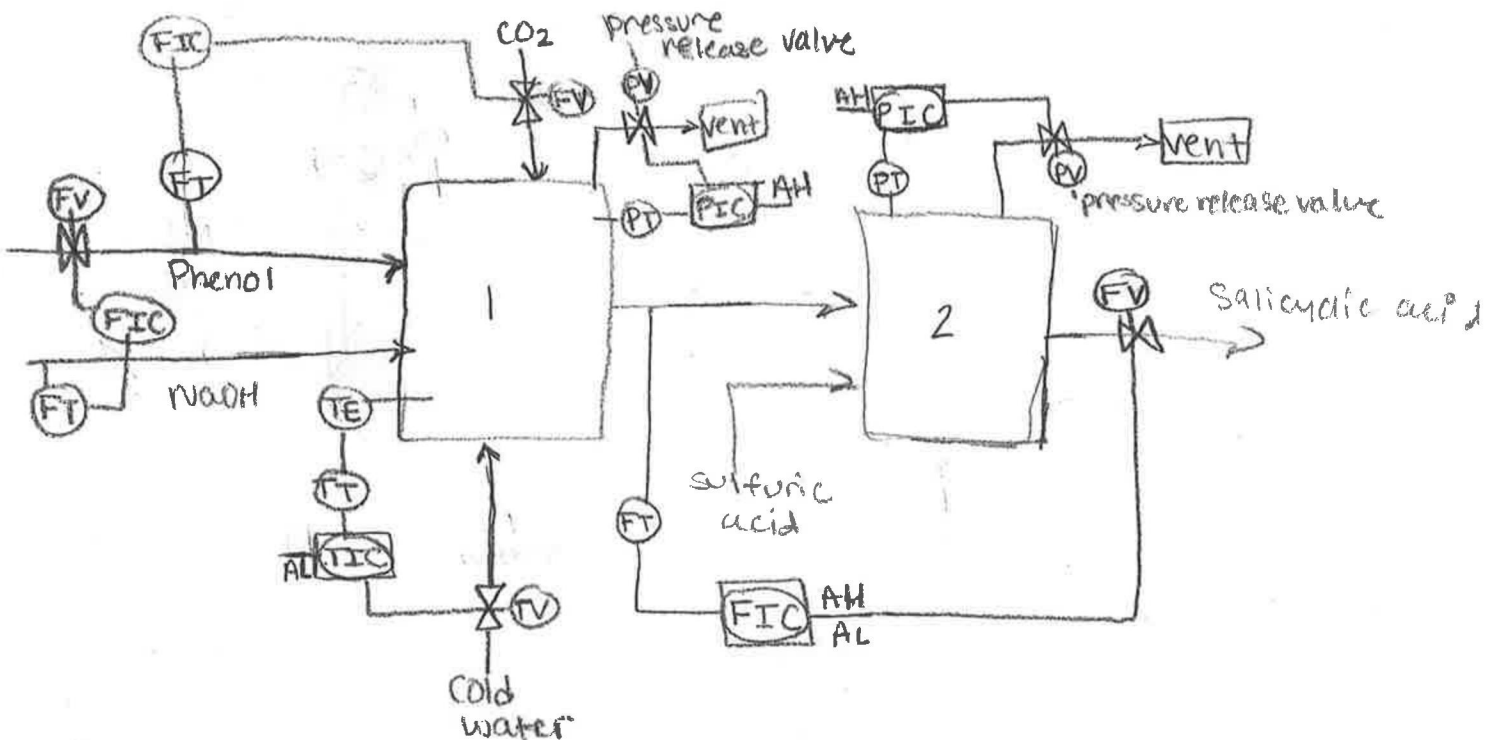
22 35 / 50

- **0 pts** Correct
- **15 pts** Mistake (part a)
- ✓ - **15 pts** Mistake (part b)
- **15 pts** Mistake (part c)
- **50 pts** wrong
- **5 pts** calculation error

Zainab Ahmed
Name

1. (50 points) Salicylic acid is usually manufactured on a commercial scale by two step process. In first step, liquid phenol and aqueous sodium hydroxide are reacted with gaseous carbon dioxide under superatmospheric pressure (150 atm) at a temperature of about 180 C. This reaction is exothermic. As a second step, the phenolic salt product obtained from first step is acidified with sulfuric acid in a second reactor to form salicylic acid. Devise a control system for this process and draw up a piping and instrumentation diagram. Consider following points;

- Constant feed temperature.
- 300 moles/hour salicylic acid production rate.
- 85% of salicylic acid composition (molar based) in the product stream.
- Maximum flow rate of cold water is 8000 gallons per hour.
- Second reactor can operate up to 10 atm and second reaction does not generate heat.



2. (50 points) Schedule 40 carbon steel pipe (inside diameter 4.026 in and outside diameter 4.5 in) used to carry wastewater was corroded to half of the original value over 3 years and need to be changed. Which of the following material can be used to replace the pipe if the expected future life of the plant is 8 years? Explain your choice.

a. Schedule 40 carbon steel pipe which needs renewal in 3 years.

b. Thicker pipe, schedule 80, outside diameter 114.3mm (4.5 in), inside diameter 97.2 mm (3.826 in)

c. Stainless steel pipe which will not corrode.

Estimates cost for per unit length pipe is; schedule 40 carbon steel \$6, schedule 80 carbon steel \$8.9, stainless steel schedule 40 \$25.30. Installation and fitting for all materials is \$17.5 per unit length. The downtime required to replace the pipe does not result in a loss of production.

Sch 40 8 yrs lifecycle

a) $(4.5 \text{ in} - 4.026 \text{ in}) / 2 = 0.237 \text{ in thickness}$

$0.237 \text{ in} / (2 \times 3 \text{ yrs}) = 0.0395 \text{ ipy} \rightarrow 13 \text{ times used}$

b) Sch 80

$(4.5 \text{ in} - 3.826 \text{ in}) / 2 = 0.337 \text{ in}$

$0.337 \text{ in} / (2 \times 1 \text{ yrs}) = 0.0337 \text{ ipy}$

$\rightarrow 8.53 \text{ yrs} \rightarrow 1 \text{ time use}$

c) SS no corrosion $\rightarrow 1 \text{ time use}$

Cost

Sch 40 = \$6 $\rightarrow (\$6 + \$17.5) \times 3 = \$70.5$

Sch 80 = \$8.9 $\rightarrow (\$8.9 + 17.5) \times 1 = \boxed{\$26.4}$ cheapest

SS = \$25.30 $\rightarrow (\$25.30 + 17.5) \times 1 = \42.8

Installation = \$17.5 / length

The thicker pipe, schedule 80, should be used to replace the pipe if the expected future life of the plant is 8 years as it will be the cheapest expense of \$26.4/lifecycle.